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Research on the distribution and control mechanism of evapotranspiration in the Nanxiaohegou watershed based on an improved S–W model

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Abstract

Accurate quantification of evapotranspiration (*ET*) and its components, and identifying their controlling factors, were beneficial for the rational evaluation, planning, and management of regional water resources. Based on continuous long-term observation data and field experiments conducted from 2016 to 2020 in the Nanxiaohegou watershed, a typical small watershed for water and soil conservation on the Loess Plateau, an improved Shuttleworth-Wallace (S-W) model was used to simulated the dynamic changes of *ET* and its components in typical plantation land. Additionally, we analyzed the coupling relationships between plant transpiration (*T*), soil evaporation (*E*), and control factors using Structural Equation Modeling (SEM). The results showed that: (1) The improved S–W model had good applicability for evaluating *ET* and its components in Nanxiaohe Watershed. The threshold value of soil surface resistance (r_s^s) was 50–2500 $s \cdot m^{-1}$, exhibiting an exponential relationship with the empirical function of soil surface water content (θ); moreover, the larger the sand content in soil particles, the greater the slope of the line. (2) *ET* ranged from 276.76 mm to 402.86 mm in typical plantation land, with multi-year average contributions of *T* and *E* accounting for 51.6% and 48.4% of *ET*, respectively. The patterns of *ET*, *T*, and *E* showed no significant monthly changes but drastic fluctuations during the day. The fluctuation trends of T and E largely reflected annual precipitation patterns but lagged behind rainfall. (3) SEM analysis revealed that net radiation (R_n), air temperature (T_a), and θ had the most significant impacts on *ET*, with R_n having the largest impact on *T* (total impact of 0.614) and T_a having the greatest impact on *E* (total impact of 0.426). *T* was positively correlated with *E*, with a contribution coefficient reaching 0.503. Evaluating *ET* and its components based on an improved S–W model established a foundation for deeply revealing ecological and hydrological processes in arid and semiarid regions.

Keywords: Evaporative components; Improve the S-W model; SEM model; Evaporation resistance; Loess Plateau; Ecological hydrological processes

1. Introduction

Evapotranspiration (*ET*) is an indispensable part of the hydrological cycle[1], playing a significant role in the water and energy balance of ecosystems. Particularly in the Loess Plateau region, *ET* accounts for more than 80% of the water loss in the system[2]. As a key factor in plant water requirements and water resource evaluation, *ET* is also an important indicator for drought monitoring. Its two components, plant transpiration (*T*) and soil evaporation (*E*), involve different biological and physical processes and have distinct dominant factors[3]. Therefore, accurately simulating *ET* and its components and studying the control mechanisms of evapotranspiration distribution are significant for the efficient planning, management, and application of water resources in the loess gully region.

Currently, *ET* assessment methods mainly include stable isotopes, sap flow, and eddy covariance techniques[4], which have the advantage of direct measurement without being constrained by hypothetical conditions[5]. However, they are limited in practical application because of high economic costs, low observation frequencies, difficulties in instrument maintenance, and narrow applicability. Hence, mathematical modeling is the primary method for *ET* assessment, with the Penman-Monteith model (P-M model) [6-7] and the Shuttleworth-Wallace model (S-W model) [8-9] being widely used for their simplicity, adaptability, and cost-effectiveness, especially in areas with complex geomorphological conditions. However, the P-M model cannot separate the components of evapotranspiration, whereas the S-W model is better due to its ability to capture physical processes, fewer parameters with clear physical meanings, and higher simulation accuracy[10]. The improved S-W model can address the issue of calculating soil surface resistance for different soil textures[11-12] and has been proven effective in *ET* separation applications in arid and semi-arid areas.

Regarding the control mechanisms of *ET* distribution and the interrelationships among its factors, scholars worldwide often analyze these through comparing pairwise correlation coefficients or using traditional statistical methods like multiple regression[12,14-15]. These methods only analyze the relationship between one or more independent variables and one dependent variable, failing to assess multiple independent variables' impact on multiple dependent variables. They also often overlook the autocorrelation between independent and dependent variables, leading to inconsistent or even opposite results[16-17]. In recent years, Structural Equation Modeling (SEM) has emerged as an effective method to address above issues. SEM not only fully considers the correlation between independent variables but also distinguishes the impact of independent variables on dependent variables, which has been widely applied in economics and ecology[18], although its application in analyzing the control factors of evapotranspiration is still limited.

As a typical prototype observation watershed in the loess gully area, this study uses long-term continuous observation data and field experiments from the Nanxiaohegou Watershed to quantify and dynamically evaluate *ET* and its components through the improved S-W model. The SEM model is used to analyze the control mechanisms, clarifying the response process between *ET* and its components and control factors. This provides a reference for estimating plant water requirements, improving water use potential, and conducting drought monitoring and assessment research in the loess gully area of the Loess Plateau.

2. Materials and Methods

2.1 Overview of the study area

The study area was located in the Nanxiaohegou Watershed, within the Xifeng District of Qingyang City, Gansu Province (35°40′43"~35°44′58"N, 107°30′8"~107°37′52"E), covering an area of 38.7 km^2. This watershed, established in 1951 by the Xifeng Station of the Loess Plateau Soil and Water Conservation Experiment, served as a typical prototype observation site for the gully region of the Loess Plateau (Figure 1). It had the distinctive topographical features of the Loess Plateau gully region, including tablelands, ridges, and valleys. The geological structure of the watershed was simple, with sandy loam soil that was prone to erosion and water loss. In recent years, with the implementation of large-scale soil and water conservation efforts in the Loess Plateau region, the vegetation coverage in the watershed had significantly increased. The main vegetation types included black locust (*Robinia pseudoacacia*), Oriental arborvitae (*Platycladus orientalis*), Chinese pine (*Pinus tabulaeformis*), Siberian apricot (*Armeniaca sibirica*), and alfalfa (*Medicago sativa*).

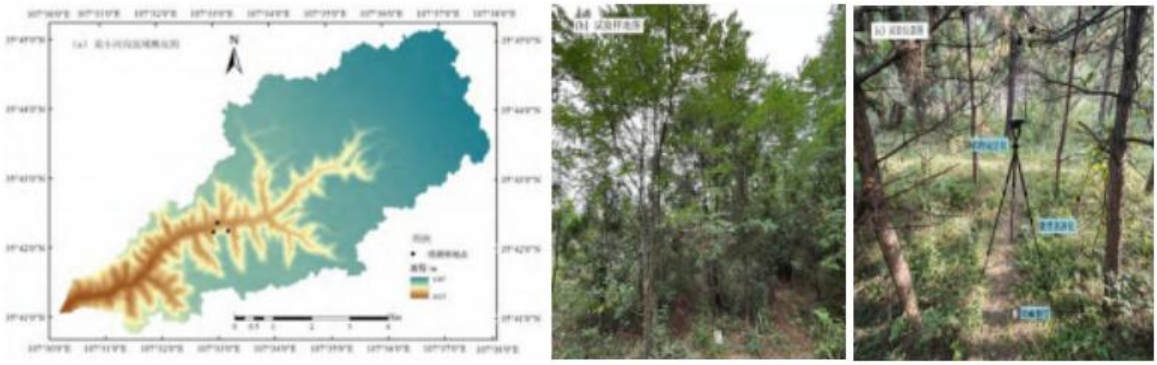


Fig 1. Schematic diagram of study area and plot and partial test map

2.2 Data sources and processing

Three sample plots with consistent tree ages and good growth were selected for the mixed plantations of black locust, Oriental arborvitae, and Chinese pine. Field experiments and observations were conducted within these plots from May to October in the years 2016, 2017, 2018,

2019, and 2020. The observations primarily included meteorological elements, soil moisture content, soil evaporation, and vegetation leaf area index. The plant parameters were represented by the average values of the three tree species, and the specific conditions of the tree species are presented in Table 1.

Table 1. Basic information of the test site

树种	树高/cm	胸径/cm	冠幅/cm	土质	地形	坡位	坡度/（°）
刺槐	564.11±29.60	15.91±2.42	293.11±15.20	砂壤土	阶地	中	26
侧柏	499.43±27.91	17.44±3.73	211.42±11.63	砂壤土	阶地	中	22
油松	446.21±24.23	15.41±1.90	191.60±12.31	砂壤土	阶地	中	21

2.2.1 Meteorological data

Meteorological data were measured by three small field weather stations (Watch dog series 2000, China). The instruments, powered by solar energy, recorded indicators such as wind speed, wind direction, air temperature, humidity, total radiation, and precipitation. The weather stations were installed in open areas near the three sample plots, with an observation interval set at 1 hour. The observation results were converted into daily-scale data for subsequent analysis and calculation. To avoid missing or distorted data, the weather stations were checked and maintained approximately every week. Any missing data were rare and were replaced by data from the nearest weather station.

2.2.2 Soil moisture content

Soil moisture content was measured using TDR (Trime TDR, Germany) probes. Experiments were conducted on the three sample plots, with two measurement points at each plot, resulting in a total of six TDR probes. The probes were buried at a location equivalent to 0.7 times the average radius of the canopy. During the installation of the probes, efforts were made to minimize disturbance to the original soil of the plantations. The TDR probes measured soil moisture at depths of 0 to 2 meters, with measurement intervals of 10 cm from 0 to 1 meter and 20 cm from 1 to 2 meters. The observation interval ranged from 1 to 3 days, and the average values were calculated. To reduce errors, observations were conducted between 8:00 and 11:00 AM each day, with three measurements at each depth in each probe. The soil moisture content of each soil layer was determined by taking the average of the three observations. Additional measurements of soil moisture content were conducted after each rainfall event.

2.2.3 Soil evaporation

Soil evaporation was measured using micro-lysimeters. The observations were conducted on the three sample plots, with three measurement points at each plot, resulting in a total of nine PVC tubes with a length of 25 cm and a diameter of 16 cm. During installation, the PVC tubes were first driven vertically into the soil using a soil tamper, and then the original soil was carefully removed, sealed with gauze at the bottom, and placed back in position, ensuring that the soil surface inside

the micro-lysimeter was level with the tube opening. Vegetation on the soil inside the PVC tubes was removed. A wire handle was attached to the top for easy subsequent observations. Observations were conducted at 18:00 every day, with an interval of 1 to 3 days. Actual evapotranspiration was calculated using the weighing method with an electronic scale of 0.01 g precision. The difference between two weighings was considered as the soil evaporation rate during a certain period. To ensure the measurement accuracy of the micro-lysimeters, the soil should be changed every 20 days, and the mass of the PVC tube and the soil inside should be consistent with the mass of the previous observation when changing the soil.

2.2.4 Leaf area index

Leaf Area Index (LAI) was measured using a plant canopy observation system (Winscanopy 2006, Canada). The measurements were conducted on the three sample plots, with two fixed observation points at each plot. After leveling the tripod, a "fisheye" camera was used to photograph three to five representative trees. The photos were processed in the Winscanopy software to obtain the LAI. The observation interval was 4 to 7 days, and the average value was calculated.

2.3 Research Methods

2.3.1 S-W model

The Shuttleworth-Wallace model (S-W model) assumed a uniform plant canopy coverage and introduced canopy resistance and soil resistance parameters. It considered evapotranspiration as originating from two "sources": soil evaporation and plant transpiration. The total water vapor source could be represented by the latent heat flux λ*E*, and its calculation formula was as follows [10]:

$$\lambda ET = \lambda E + \lambda T = C_s PM_s + C_c PM_c \quad (1)$$

in the formula,

$$E = PM_s = \frac{\Delta R + \dfrac{\rho C_p D - \Delta r_a^c \left(R - R_s\right)}{r_a^a + r_a^s}}{\Delta + \gamma\left(1 + \dfrac{r_s^s}{r_a^a + r_a^s}\right)} \quad (2)$$

$$T = PM_C = \frac{\Delta R + \dfrac{\rho C_p D - \Delta r_a^c R_S}{r_a^a + r_a^c}}{\Delta + \gamma\left(1 + \dfrac{r_s^c}{r_a^a + r_a^c}\right)} \quad (3)$$

In the formula: *ET* represented the simulated evapotranspiration rate (mm d^{-1}); λ was the latent heat of vaporization of water (MJ kg^{-1}); PM_s was the simulated soil evaporation rate (mm·d^{-1}); PM_c was the simulated plant transpiration rate (mm·d^{-1}); C_s and C_c were the proportionality coefficients

for soil evaporation and plant transpiration, respectively; r_s^s was the soil surface resistance ($s \cdot m^{-1}$); r_a^s was the aerodynamic resistance from the soil surface to the canopy height for latent heat ($s \cdot m^{-1}$); r_s^c was the stomatal resistance of the plant canopy ($s \cdot m^{-1}$); r_a^c was the aerodynamic resistance from the leaf surface to the average height of the canopy ($s \cdot m^{-1}$); r_a^a was the aerodynamic resistance from the canopy height to the reference height for latent heat ($s \cdot m^{-1}$); Δ was the slope of the saturation vapor pressure curve with temperature ($kPa \cdot ℃^{-1}$); ρ was the air density (1.29 $kg \cdot m^{-3}$); C_p was the specific heat of air at constant pressure [1.03×10^{-3} J $\cdot (kg \cdot ℃)^{-1}$]; D was the air saturation vapor pressure difference (kPa); γ was the psychrometric constant (0.67 $hPa \cdot k^{-1}$).

In the formula: R represented the radiation absorbed by the canopy ($W \cdot m^{-2}$), and R_S represented the radiation absorbed by the soil ($W \cdot m^{-2}$). The calculation formula was as follows:

$$R = R_n - G \tag{4}$$

$$R_s = R_{ns} - G \tag{5}$$

$$R_{ns} = R_n exp\left(-C\mathrm{LAI}\right) \tag{6}$$

In the formula: C was the extinction coefficient, with a value of 0.60; G was the soil heat flux ($W \cdot m^{-2}$); R_n was the net radiation ($W \cdot m^{-2}$), and R_{ns} was the net radiation absorbed by the soil ($W \cdot m^{-2}$), which was calculated based on the LAI extinction principle [19].

2.3.2 Structural Equation Modeling (SEM)

Structural Equation Modeling (SEM) was a method based on the covariance matrix that simultaneously analyzed the causal relationships between multiple variables. It integrated factor analysis and path analysis, allowing for the establishment, estimation, and testing of causal relationships in models. SEM could analyze the total, direct, and indirect effects of independent variables on dependent variables, making it suitable for studying the inter relationships between multiple variables. Typically, SEM used maximum likelihood estimation for parameter estimation, and the fit function of maximum likelihood estimation conformed to the χ^2 distribution. Therefore, in SEM, the χ^2 test was often used to evaluate model performance. The software Amos 24 was used to run SEM, and the model was adjusted and revised based on the test results to analyze the control mechanisms of plant transpiration (T) and soil evaporation (E).

2.4 Calculation of model improvement parameters

The S-W model included a large number of parameters, in addition to environmental parameters, the core of its formula was the calculation and validation of five resistance parameters: r_s^s, r_a^s, r_s^c, r_a^c, and r_a^a. Based on the research findings of other scholars and combined with the actual conditions of the study area, the research methods for the three parameters r_a^a, r_a^c, and r_a^s were relatively unified. This study followed the original model method and still used the formulas

from reference [10] for calculation. Additionally, the two parameters r_s^c and r_s^s adopted the methods recommended by other scholars, and the calculation methods were as follows.

2.4.1 Vegetation canopy stomatal resistance(r_s^c)

r_s^c was expressed as a function of solar radiation, water vapor deficit, air temperature, and soil moisture content, according to the Jarvis method [21], namely:

$$r_s^c = \frac{r_{smin}}{LAI}\frac{F_1}{F_2F_3F_4} \tag{7}$$

$$F_1 = \frac{1+f}{f+\frac{r_{smin}}{r_{smax}}} \tag{8}$$

$$f = \frac{0.55R_{si}}{R_{sl}}\frac{2}{LAI} \tag{9}$$

$$F_2 = 1 - 0.0025D \tag{10}$$

$$F_3 = 1 - 0.0016(298 - T) \tag{11}$$

$$F_4 = \begin{cases} 1 & \theta r > \theta wp \\ \dfrac{\theta r - \theta wp}{\theta fc - \theta wp} & \theta fc \le \theta r \le \theta wp \\ 0 & \theta r < \theta wp \end{cases} \tag{12}$$

In the formula: r_{smin} represented the minimum stomatal resistance of the plant canopy ($s \cdot m^{-1}$), and r_{smax} represented the maximum stomatal resistance of the plant canopy ($s \cdot m^{-1}$). According to the research results of Cui Jianguo [22], they were taken as 170 $s \cdot cm^{-1}$ and 1400 $s \cdot cm^{-1}$, respectively; F_i (1, 2, 3, 4) denoted the stress functions for specific environmental variables; R_{si} was the shortwave radiation absorbed by the plant ($W \cdot m^{-2}$); R_{sl} was the solar radiation at the onset of photosynthesis ($W \cdot m^{-2}$); T_a was the air temperature (K); θ_r represented the soil moisture content at the plant root zone ($cm^3 \cdot cm^{-3}$, in this paper, the soil moisture content at a depth of 100~120 cm was used as a substitute); θ_{fc} was the field capacity ($cm^3 \cdot cm^{-3}$, usually taken as $0.75\theta_{sat}$); θ_{wp} was the plant wilting coefficient (0.05 $cm^3 \cdot cm^{-3}$);

2.4.2 Soil surface resistance(r_s^s)

r_s^s adjusted the movement of water vapor from the interior of the soil to the soil surface and had a strong dependence on the moisture content of the upper soil layer. Studies have shown tha (13)
was a functional relationship between r_s^s and the soil surface moisture content θ [23]. That was,

$$r_s^s = a_1(\frac{\theta_{sat}}{\theta})^{b_1} + c_1$$

In the formula: θ represented the soil surface moisture content ($cm^3 \cdot cm^{-3}$). Due to the complexity of the soil surface, the soil moisture content at a depth of 10 cm was used as a substitute in this paper. θ_{sat} was the soil saturated moisture content (0.41 $cm^3 \cdot cm^{-3}$), and a_1, b_1, c_1 were parameters that needed to be calibrated in the model.

2.5 Model evaluation methodology

The purpose of model evaluation was to determine whether the model met the research requirements. In this paper, the Root Mean Square Error (RMSE), Coefficient of Determination (R^2), and Nash-Sutcliffe Efficiency Coefficient (NS) were used to evaluate the accuracy of the model simulation. The calculation formulas and evaluation criteria were detailed in reference [2].

3. Results

3.1 Improved S-W Model Application

3.1.1 Model Verification

Using meteorological data and observation data from June to October of 2016, 2017, and 2018 for parameter calibration, and data from 2019 and 2020 for model verification. Based on the original model's three parameters r_a^a, r_a^c, r_a^s and the substituted r_s^c, r_s^s two parameters' calculations, substituting into Formula (2), soil evaporation simulation test was conducted, and the simulation results were shown in Figure 2.

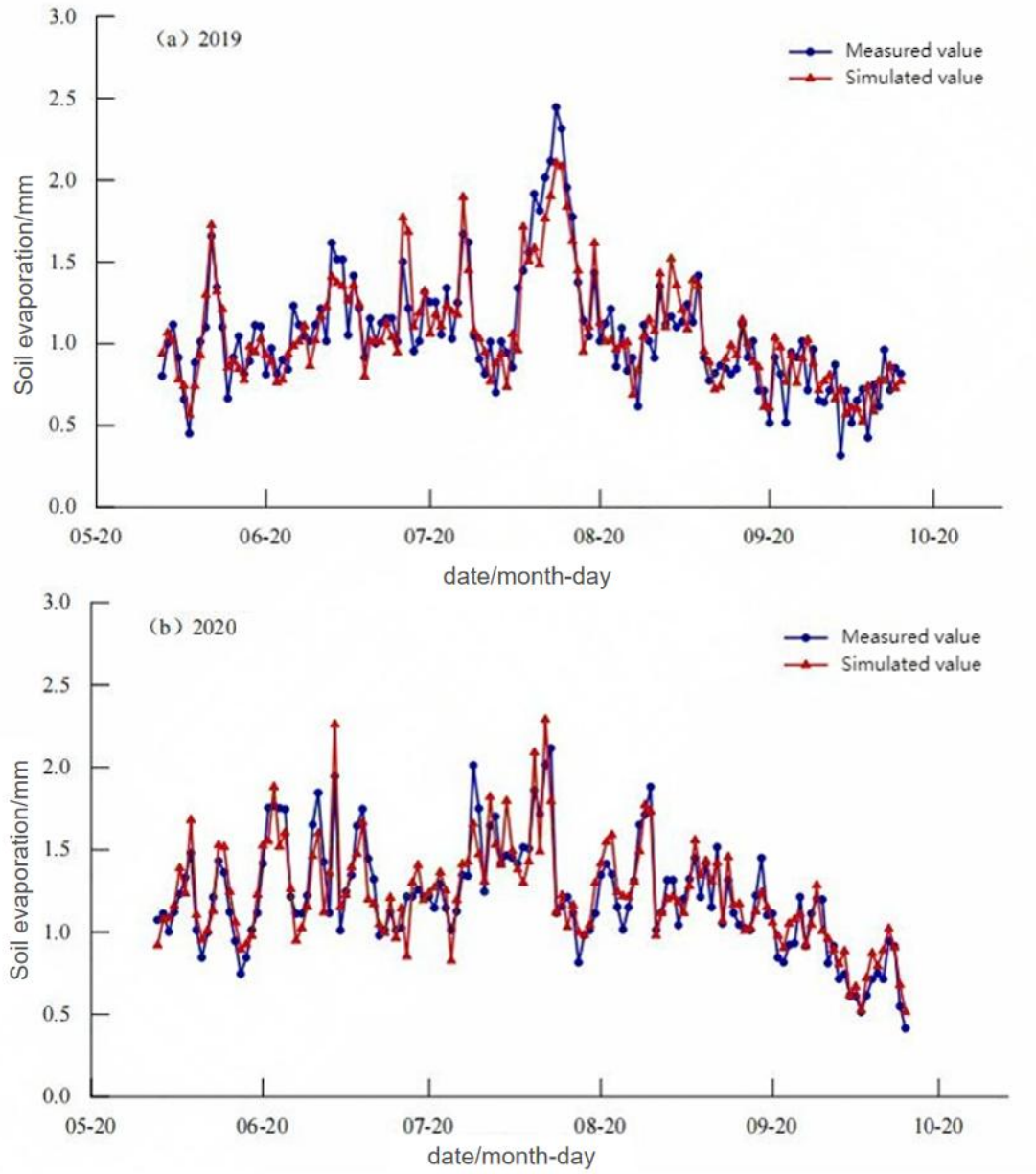


Fig 2. Comparison between simulated and measured

From Figure 2, it could be seen that the improved S-W model performs well in simulating soil evaporation, with small differences between measured and simulated values. The accuracy assessment indicators RMSE, R^2, NS for 2019 and 2020 were 0.16 mm·d^{-1}, 0.80, 0.81 and 0.13 mm·d^{-1}, R^2=0.82, NS=0.84 respectively. All evaluation indicator results were within reasonable ranges, indicating that the parameter calculations in the model were reasonable, and the improved S-W model was suitable for simulating typical artificial forests in the Nan Xiaohe River Basin.

3.1.2 Dynamic Simulation of Evapotranspiration and Its Components

Daily dynamic simulations of evapotranspiration and its components for typical artificial forests in the Nan Xiaohe River Basin were performed (Figure 3).

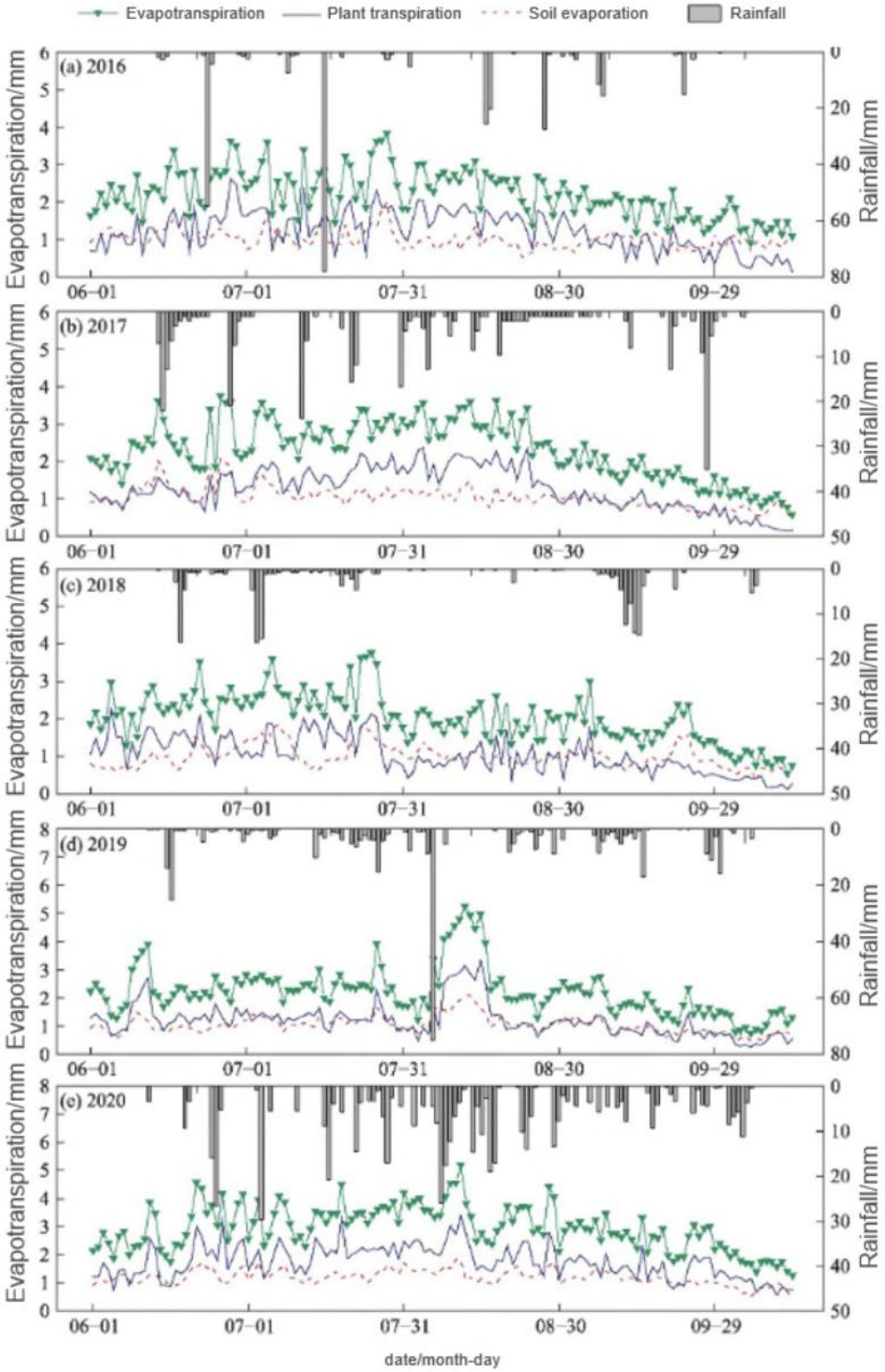


Fig 3. Dynamic changes of evapotranspiration, plant transpiration and soil evaporation with rainfall in the typical plantation

From Figure 3, we saw that in five experimental years, the variation of evaporation and transpiration of typical artificial forest land, plant transpiration, and soil evaporation showed no obvious monthly changes but had intense daily fluctuations. The trends of plant transpiration and soil evaporation were basically consistent. The range of simulated evaporation change was 276.76~402.86 mm, among which soil evaporation was 134.26~167.08 mm, with an average daily value of 0.34~2.10 mm/d^{-1}; Plant transpiration was 142.26~239.12 mm, with an average daily value of 0.14~3.40 mm/d^{-1}; The maximum values of evaporation, plant transpiration, and soil evaporation generally appeared in July, decreased significantly after mid-September, and even showed phenomena where plant transpiration approached or was lower than soil evaporation. Statistics showed that during the growing season, the ratio of daily plant transpiration to daily evaporation (*T:ET*) was 41.5%~59.8%, while the ratio of daily soil evaporation to daily evaporation (*E:ET*) was 40.2%~58.5%.

Evaporation, plant transpiration, and soil evaporation responded to rainfall with annual variations showing similar trends, but there was a certain lag relative to precipitation events. The amount of evaporation, plant transpiration, soil evaporation, and rainfall all showed wet year (2020) > normal year (2016, 2017) > dry year (2018). Affected by heavy rainfalls (June 17, 2016, July 13, 2017, June 10, 20, 2018, July 23, 2019, August 6, 2020, August 8, etc.), evaporation, plant transpiration, and soil evaporation showed time lags behind precipitation peaks.

3.2 Factors Affecting Plant Transpiration and Soil Evaporation

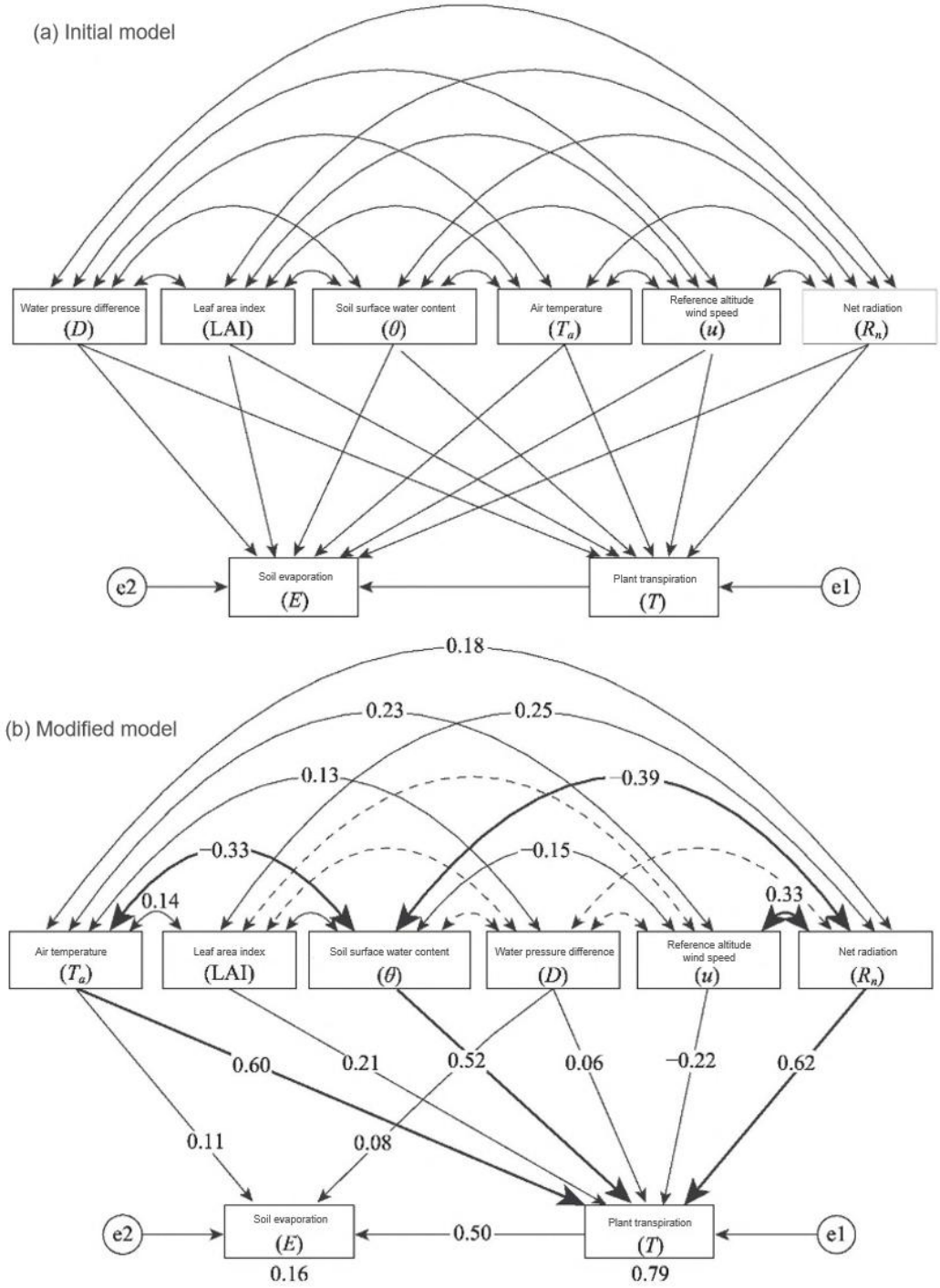
Studies had shown[8,24] that the main factors affecting plant transpiration and soil evaporation included leaf area index, soil moisture content, air saturation water vapor pressure deficit, etc. This paper combined the S-W model calculation parameters to analyze the control factors of plant transpiration and soil evaporation, taking leaf area index, soil moisture content, air saturation water vapor pressure deficit, temperature, solar altitude wind speed, and net radiation as six independent variables, and plant transpiration and soil evaporation as two dependent variables, using data from 2016-2020 (a total of 681 groups), based on structural equation modeling (SEM) to establish the initial hypothesis model (Figure 4a).

3.2.1 Model Fit Test, Adjustment, and Correction

(1) Initial Testing and Analysis

After the initial operation of the model, it was found that the P-values for soil surface layer moisture content, leaf area index, reference height wind speed, and net radiation effect on soil evaporation were 0.212, 0.297, 0.884, and 0.872 respectively, all failing to reach the 0.05 significance level test. At the same time, considering the interrelationships among the control factors, paths with smaller path coefficients that did not reach the significant level were removed,

resulting in the corrected model (Figure 4b). The standardized coefficients and significance levels of the initial and corrected models were listed in Table 2.



注：直线箭头表示箭头起始点变量对所指变量有直接影响，直线上数值表示标准化路径系数；曲线双向箭头表示变量间有相关关系，曲线上数值表示相关系数；方框下方数值表示被变量解释比例；线的粗细表示关系强弱，虚线表示相关关系不显著；e1、e2 表示残差变异项。

Fig 4. Structural equation path diagram of control factors with plant transpiration and soil water evaporation

Table 2. Standardization coefficients and significance among control factors

初始模型参数估计							修正模型参数估计						
变量	关系	变量	*Std*	*S.E.*	*C.R.*	*P*	变量	关系	变量	*Std*	*S.E.*	*C.R.*	*P*
T	←	*u*	-0.218	0.042	-9.378	***	*T*	←	*u*	-0.219	0.042	-9.378	***
T	←	*Ta*	0.588	0.003	25.195	***	*T*	←	*Ta*	0.591	0.003	25.195	***
T	←	*θ*	0.513	1.286	20.429	***	*T*	←	*θ*	0.517	1.286	20.429	***
T	←	LAI	0.205	0.015	8.831	***	*T*	←	LAI	0.206	0.015	8.831	***
T	←	*Rn*	0.609	0.000	24.78	***	*T*	←	*Rn*	0.614	0.000	24.78	***
T	←	*D*	0.064	0.024	2.952	0.003	*T*	←	*D*	0.064	0.024	2.952	0.003
E	←	*θ*	-0.057	1.132	-1.249	0.212	*E*	←	*T*	0.514	0.018	13.77	***
E	←	*Ta*	0.098	0.003	2.104	0.035	*E*	←	*D*	0.083	0.017	2.669	0.008
E	←	LAI	-0.037	0.011	-1.044	0.297	*E*	←	*Ta*	0.110	0.003	2.784	0.005
E	←	*T*	0.535	0.027	9.758	***	*Ta*	↔	*D*	0.130	0.087	3.629	***
E	←	*D*	0.084	0.017	2.697	0.007	*Ta*	↔	*Rn*	0.181	7.113	4.687	***
E	←	*u*	0.005	0.031	0.146	0.884	*u*	↔	*Rn*	0.334	0.553	8.502	***
E	←	*Rn*	-0.008	0.000	-0.161	0.872	*u*	↔	*θ*	-0.146	0.000	-4.02	***
θ	↔	LAI	-0.346	0.000	-8.522	***	*u*	↔	*Ta*	0.231	0.058	5.96	***
θ	↔	*D*	-0.013	0.000	-0.336	0.737	*θ*	↔	LAI	-0.351	0.000	-8.711	***
θ	↔	*Ta*	-0.33	0.002	-8.181	***	*θ*	↔	*Ta*	-0.332	0.002	-8.279	***
θ	↔	*u*	-0.133	0.000	-3.431	***	*θ*	↔	*Rn*	-0.387	0.020	-9.448	***
θ	↔	*Rn*	-0.383	0.020	-9.317	***	LAI	↔	*Ta*	0.137	0.158	3.644	***
LAI	↔	*D*	0.081	0.020	2.114	0.035	LAI	↔	*Rn*	0.248	1.466	6.626	***
LAI	↔	*Ta*	0.138	0.163	3.573	***							
LAI	↔	*Rn*	0.235	1.539	5.976	***							
u	↔	LAI	-0.039	0.012	-1.022	0.307							
u	↔	*D*	-0.036	0.007	-0.946	0.344							
u	↔	*Ta*	0.221	0.059	5.627	***							
u	↔	*Rn*	0.326	0.566	8.076	***							
D	↔	*Ta*	0.13	0.093	3.369	***							
D	↔	*Rn*	0.028	0.859	0.739	0.460							
Ta	↔	*Rn*	0.182	7.144	4.658	***							

注：*Std* 为标准化系数；*S.E.*为标准误差；*C.R.*为临界比；*P* 为显著性水平；***为 *P*<0.001，相关性显著；当 *P*>0.001，显示具体数值；←为自变量指向因变量；↔为主因子之间相互关系；LAI 为叶面积指数；*θ* 为土壤表层含水量；*u* 参照高度风速；*D* 为饱和水汽压差；*Ta* 为气温；*Rn* 为净辐射。

(2) Goodness-of-fit test

As could be seen from the goodness-of-fit indicators of the modified model (Table 3), both the absolute fit indicator and the incremental fit indicator met the standard, indicating good fit.

Table 3. SEM model fit test

检验统计量	参数	判别标准		分析结果
		原则	依据	
绝对拟合指标	卡方自由度比值（χ^2:DF）	1<（χ^2:DF）<3	越接近 1 越好	1.026
	绝对拟合指数（GFI）	0.9<GFI<1	越接近 1 越好	0.997
	近似误差均方根（RMSEA）	0<RMSEA	越低越好	0.006
相对拟合指标	比较适配指数（CFI）	0.9<CFI<1	越接近 1 越好	0.999
	规准适配指数（NFI）	0.9<NFI<1	越接近 1 越好	0.994

3.2.2 Control Factors of Plant Transpiration and Soil Evaporation

To better compare the importance of each control factor in the model, the results of Figure 4 were organized into Table 4. After correction, the SEM model showed an overall explanatory rate of 79.1% for plant transpiration and 61.3% for soil evaporation. As indicated in Table 4, plant transpiration positively affected soil evaporation, meaning soil evaporation increased with plant transpiration, with a path coefficient of 0.503. Among all control factors, net radiation had the greatest impact on plant transpiration, with a total influence of 0.614. Other factors influencing plant transpiration, ranked from highest to lowest, were air temperature, soil moisture content, leaf area index, and wind speed at the reference height. All these factors directly affected plant

transpiration. Air temperature exerted the most significant influence on soil evaporation, with a total impact of 0.426. Besides vapor pressure deficit and air temperature, other factors indirectly affected soil evaporation.

Table 4. Influence coefficients of control factors on plant transpiration and soil water evaporation

		饱和水汽压差	净辐射	气温	叶面积指数	土壤表层含水量	参照高度风速	植物蒸腾
总影响	植物蒸腾	0.064	0.614	0.591	0.206	0.517	-0.219	-
	土壤蒸发	0.116	0.309	0.426	0.104	0.260	-0.110	0.503
直接影响	植物蒸腾	0.064	0.614	0.591	0.206	0.517	-0.219	-
	土壤蒸发	0.083	-	0.128	-	-	-	0.503
间接影响	植物蒸腾	-	-	-	-	-	-	-
	土壤蒸发	0.032	0.309	0.298	0.104	0.260	-0.110	-

注：“-”表示几乎无影响。

Note: "-" indicates little or no impact.

The SEM model also displayed the correlation among six control factors (Table 2). Among them, the soil surface moisture content had the highest correlation with the remaining five control factors, and their correlations were negative; the largest correlation was with net radiation, with a correlation coefficient reaching -0.387; followed by the leaf area index, with a correlation coefficient of -0.351. In positive correlations, the correlation between the wind speed at the light transmission height and net radiation was the highest, at 0.334; followed by the correlation between the leaf area index and net radiation, with a correlation coefficient of 0.248. The above relationships among the control factors were reasonable.

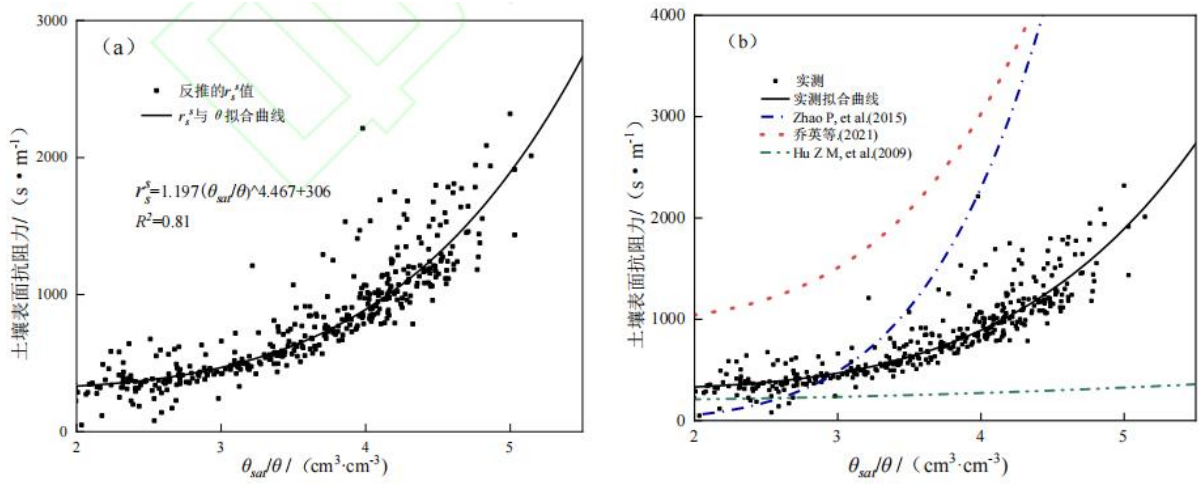
4. Discussions and Conclusion

4.1 Discussions

4.1.1 the parameter calibration of soil surface resistance (r_s^s)

Due to the differences in soil types and textures across various regions, r_s^s numerous factors are involved, making it difficult to determine the parameters through a unified method, and they are often estimated using empirical formulas. Previous studies indicate that the types of empirical formulas r_a^s related to θ include linear forms[25], exponential forms[23], and power functions[26].This paper determines the functional type of the r_s^s relationship with θ in the Nanxiaohe Gully watershed to be exponential (Figure 5a), r_s^s increasing as θ decreases. This is because as θ lowers, the difference in water vapor pressure inside and outside the soil decreases, reducing the rate of water vapor transfer, which leads to an r_s^s increase in [27]. Studies by Zhao et al.[23], Qiao et al.[24], and Hu et al.[28] also conclude that the functional type of the relationship r_s^s with θ is exponential. Through comparative analysis (Figure 5b), it is found that the more sand content in the soil texture of the study area, the more pronounced the exponential form of the relationship between r_s^s and θ, and the greater the sand content, the larger the power index. This can provide a reference for

empirical formula estimation in other regions.



注：θ_{sat}/θ 为土壤饱和含水量与土壤表层含水量的比值

Fig 5. Correlation between soil surface resistance and soil moisture content

4.1.2 Evapotranspiration and component ratios

Evapotranspiration is an important process in the water balance and energy equilibrium of ecosystems, closely related to ecosystem productivity. The stronger the plant transpiration, the more beneficial it is for the improvement of the ecological environment. The less soil evaporation, the higher the water use efficiency. Therefore, the components of evapotranspiration can reflect the characteristics of water cycling in different ecosystems. Studies have found [9,29,30] that during the growing season of plants, the ratio of plant transpiration to evapotranspiration is generally about 60% to 80%, while the ratio of plant transpiration to evapotranspiration in the study area is relatively low (an average of 51.6%), which may be due to the combined effects of local soil moisture, topography, climatic conditions, and vegetation types. The study area is located in an arid to semi-arid climate zone, with complex topography such as plateaus, slopes, and gullies, and the vegetation type is coniferous forest of artificial origin. Compared to areas with rich types and high coverage of broadleaf forests or grassland vegetation, soil evaporation consumes more, and the proportion of plant transpiration is relatively lower. It can be seen that there is still a significant room for improvement in the efficiency of water absorption by plants in the Nanxiaohe Gully watershed.

4.1.3 Based on the SEM model, the control factors of evapotranspiration and its components are analyzed.

Traditional studies on the control factors of plant transpiration and soil evaporation only analyze the impact of control factors on them, often neglecting the interrelated effects between plant transpiration and soil evaporation, as well as among the control factors. This paper uses the SEM model to explore the impact and path contribution of six control factors, including Leaf Area Index

(LAI) and soil surface water content, on plant transpiration and soil evaporation, and analyzes the correlation among the control factors. The study shows that net radiation, air temperature, and soil surface water content are the most important control factors for evapotranspiration, with net radiation having the greatest impact on plant transpiration (total impact of 0.614), which is consistent with the research results of Liu et al.[31]. Solar radiation is the largest source of energy; the stronger the net radiation, the more sufficient the driving force it can provide for plant transpiration, leading to an increase in the amount of plant transpiration.

Related research has found that factors affecting soil evaporation can generally be divided into two major categories: meteorological factors (air temperature, saturation vapor pressure deficit, reference height wind speed, net radiation, etc.) and water supply conditions, with the latter primarily determined by soil surface water content [32]. Most scholars believe that when soil moisture is lacking, soil surface water content is the main controlling factor of soil evaporation [33]; however, some scholars consider air temperature to be the primary controlling factor of soil evaporation [34]. The conclusions of this study align with the latter, with air temperature having a total impact of 0.426 on soil evaporation, while soil surface water content has a total impact of 0.260. The reason may be that only on the basis of suitable soil surface water content, air temperature is the main controlling factor of soil evaporation. Under vegetation disturbance, suitable soil surface water content is a prerequisite for the impact of air temperature on soil evaporation; SEM model analysis also indicates that soil surface water content has the highest correlation with the other five controlling factors (including air temperature), and all controlling factors mainly exhibit an indirect impact on soil evaporation. Additionally, the study area has a smaller amplitude of daily variation in soil surface water content. Therefore, although the study area has less soil moisture, the aforementioned reasons weaken the impact of soil surface water content on soil evaporation to a certain extent. Discussing the impact process of vegetation and meteorological factors on soil evaporation, this study finds that the total impact of plant transpiration on soil evaporation reaches 0.503, which is higher than the impact of the maximum meteorological factor, air temperature, on soil evaporation. This indicates that the impact of vegetation on soil evaporation is greater than the impact of meteorological factors on soil evaporation, consistent with the research of Yang et al. [35]. Furthermore, the explanatory power of the six selected controlling factors for soil evaporation is 61.3%, with a slightly larger unexplained percentage, suggesting that compared to plant transpiration, soil evaporation may have more influencing factors and a more complex process, and the controlling factors selected for analysis in the SEM model may be slightly fewer.

4.2 Conclusion

Based on the field experimental data from the Nanxiaohe Gully watershed from 2016 to 2020, the improved S-W model is applied to quantify and dynamically simulate *ET*, *T*, and *E* for typical artificial forest land. The simulation finds that the improved S-W model has good applicability in the typical artificial forest land of the Nanxiaohe Gully watershed. The annual average *ET* for the typical artificial forest land is 320.24 mm, with *T* and *E* accounting for 51.6% and 48.4% of *ET*, respectively. The proportion of *E* is relatively high, indicating low plant water use efficiency, which shows that there is still significant room for improving plant water absorption in the study area. Additionally, based on the SEM model analysis, the control influences of multiple meteorological factors on *T* and *E*, as well as the mutual influences among the factors, are examined. Among them, R_n has the greatest impact on *T*, while T_a has the greatest impact on *E*. The effect of vegetation on *ET* is greater than that of meteorological factors on *ET*. The influence of *T* on *E* is positively correlated. Compared to *T*, the generation process and influencing factors of *E* are more complex. Due to the differences in soil type and texture, the parameters and forms of r_s^s are difficult to determine through a unified method. In this study area, the soil surface resistance (r_s^s) is analyzed. Empirical formula is

$$r_s^s = 1.197(\frac{\theta_{sat}}{\theta})^{4.467} + 306$$

Comparing other empirical formulas, it is found that the greater the content of sand particles in the soil particles, the more obvious the exponential form, and the larger the exponent, which can provide a reference for the empirical formula estimation in other regions.

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